

Split strategy as a controlled validation variable

Same data, descriptors and model grid; only the train/test logic changes.

Benchmark design

Data

clean adsorption table
cluster labels
optional topology metadata



Matched design

same targets
same descriptors
same model grid



Split families

random
geometry / chemistry
topology stress test



Model-to-audit translation

Models

Ridge, RF, HGB, MLP
main text: RF-centred
SI: all models



Consequences

accuracy loss
rank inversion
elite-list instability



Outputs

tables + figure data
PNG/PDF figures
case-study candidates

Deliverable: auditable accuracy, screening and group-level diagnostics for MOF extrapolation.